

Transactional Magnitude Calculus: Monetary Derivatives, S-Entropy Normalisation, and the Monetary Tangent Space

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Abstract

Classical analysis measures rates of change relative to calendar time. In systems driven by economic activity, however, the natural clock is not the uniform tick of the calendar but the accumulated magnitude of transactional events. We develop a rigorous calculus in which the fundamental rate operator is the *monetary derivative* $d/d\Theta$, where $\Theta(t) = \int_0^t |G(s)| ds$ is the *transaction clock*—the running integral of absolute gain-loss intensity. Replacing calendar time with transaction time subordinates every process to the underlying economic activity and eliminates the dimensionality barriers that obstruct cross-quantity comparison: when the S-entropy functional $S : \mathcal{R} \times \mathcal{X} \times \mathcal{X}^* \rightarrow [0, 100]$ is used as the scalar field, every monetary derivative $dS/d\Theta$ is measured in *S-units per S-unit*—a pure number that adds across heterogeneous quantities without conversion factors. We formalise the *monetary tangent space* $\mathcal{T}^\Theta$ at each transaction-time point, prove that it carries a natural norm inherited from the S-functional, and show that it is closed under the monetary derivative operator. For systems governed by a hierarchical gear network of nested stopping times $\{\sigma_k\}_{k \geq 1}$, we derive a *telescoping gear-ratio formula* $dS/d\Theta_K = (dS/d\Theta_1) \prod_{k=1}^{K-1} (\Theta_k/\Theta_{k+1})$ and prove that the No-Privileged-Level principle—the same algebraic laws hold at every gear depth—is an exact consequence of the S-entropy axioms. We further prove an *ergodic consistency theorem*: under a stationary ergodic return process the time-averaged monetary derivative converges almost surely to the stationary mean, and the convergence rate is $O(K^{-1/2})$ by Birkhoff’s theorem applied in transaction time. The framework provides a principled foundation for comparing rates of change of price, volume, volatility, and any other economic observable on a sin-

gle dimensionless scale, and reduces to classical calculus when transaction activity is constant.

Keywords: transaction clock, monetary derivative, S-entropy, monetary tangent space, gear network, No-Privileged-Level, Clark subordination, ergodic theory, dimensional analysis, gain-loss intensity.

Contents

1 Introduction

The calculus of variations, Newtonian mechanics, and classical stochastic analysis all share a common primitive: the calendar-time derivative d/dt . For physical systems whose natural clocking is the uniform passage of time this is irreproachable. But financial markets, biological signalling cascades, supply networks, and social contagion all possess an internal clock whose rate is itself driven by the intensity of the phenomenon being studied. A stock that trades ten thousand shares in a second has advanced further along its economic trajectory than a stock that trades ten shares over a day, even if both share the same calendar elapsed time.

This observation has antecedents. Clark (?) showed empirically that daily price changes are better described by a subordinated process: Brownian motion in transaction time, where transaction time is the accumulated volume or trade count. Mandelbrot and Taylor (?) and Monroe (?) formalised the connection between continuous local martingales and Brownian motion in a subordinated clock. Ane and Geman (?) confirmed empirically that conditioning on trade count renders intraday returns approximately Gaussian. In the mathematical

finance literature, stochastic time changes appear in the context of variance processes and activity rates (?).

Yet despite these advances, no systematic calculus has been developed that:

1. replaces the calendar-time derivative d/dt with a *monetary derivative* $d/d\Theta$ uniformly across all quantities of interest;
2. resolves the *dimensional incommensurability* problem that arises when one attempts to add rates of change of heterogeneous observables (price velocity cannot simply be added to volume velocity);
3. provides a *scale-free, addable* scalar for rates of change that respects the algebraic structure of bounded-cognition information theory.

The present paper addresses all three gaps. The key insight is that when rates of change are measured in the S-entropy scale $[0, 100]$ of ?—a bounded, receiver-relative distance-to-truth functional—and normalised by the transaction clock, the result is a pure number: S-units of distance-to-truth per S-unit of distance-to-truth. Two monetary derivatives of entirely different quantities are then directly addable, subtractable, and comparable.

The paper is organised as follows. Section ?? sets up notation and reviews the required background on graph Laplacians, Kirchhoff portfolios, and S-entropy. Section ?? defines the transaction clock and proves its basic properties. Section ?? introduces the monetary derivative operator and develops its analysis. Section ?? shows how S-entropy eliminates dimensional barriers. Section ?? constructs the monetary tangent space and establishes its normed-space structure. Section ?? derives the telescoping gear-ratio formula for hierarchical networks. Section ?? proves the No-Privileged-Level theorem in transaction time. Section ?? establishes ergodic consistency of monetary derivatives. Section ?? works three illustrative examples. Section ?? discusses implications and open questions.

2 Preliminaries

2.1 Probability and filtration

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space and $(\mathcal{F}_t)_{t \geq 0}$ a right-continuous filtration sat-

isfying the usual conditions. All processes are assumed to be $(\mathcal{F}_t)$ -adapted.

2.2 Graph Laplacian and Kirchhoff portfolio

Let $G = (V, E, A)$ be a weighted undirected graph with $m = |V|$ assets, adjacency matrix $A \in \mathbb{R}^{m \times m}$ ($A_{ij} = A_{ji} \geq 0$, $A_{ii} = 0$), degree matrix $D = \text{diag}(A\mathbf{1})$, and Laplacian $\mathbf{L} = D - A$. Since G is connected, $\mathbf{L}$ has eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_m$; λ_2 is the Fiedler (algebraic connectivity) value.

For a forecast return vector $\hat{\mu} \in \mathbb{R}^m$ with mean-centred version $\mu_c = \hat{\mu} - \bar{\hat{\mu}}\mathbf{1}$, the *Kirchhoff fixed-point portfolio* is

$$w^* = \mathbf{L}^\dagger \mu_c + \frac{1}{m} \mathbf{1}, \quad (1)$$

where $\mathbf{L}^\dagger$ is the Moore–Penrose pseudoinverse of $\mathbf{L}$ (?). The weight w^* satisfies the discrete Kirchhoff law $\mathbf{L}w^* = \mu_c$ and lies on the probability simplex Δ_m (?).

2.3 Gain-loss process

Let $R(t, \tau) \in \mathbb{R}^m$ be the vector of asset returns over the interval $[t, t + \tau]$, and let $\hat{\mu}(\tau) = \mathbb{E}[R(t, t + \tau)]$ be the corresponding forecast. The *Kirchhoff gain-loss* at calendar time t with horizon τ is

$$G(t, \tau) = (w^*(\tau))^\top [R(t, t + \tau) - \hat{\mu}(\tau)]. \quad (2)$$

When the forecast is conditionally unbiased—i.e. $\mathbb{E}[R(t, t + \tau) \mid \mathcal{F}_t] = \hat{\mu}(\tau)$ —the process $(G(t, \tau))_t$ is a martingale difference sequence with respect to $(\mathcal{F}_t)$.

2.4 S-entropy functional

We recall the S-entropy framework of ?. Let $\mathcal{X}$ be a candidate space, $\mathcal{X}^* \subseteq \mathcal{X}$ the truth space, and $\mathcal{R}$ the class of all receivers. An *S-functional* is a map

$$S : \mathcal{R} \times \mathcal{X} \times \mathcal{X}^* \rightarrow [0, 100], \quad (\mathcal{R}, x, x^*) \mapsto S_{\mathcal{R}}(x, x^*), \quad (3)$$

satisfying non-negativity, boundedness, the positive floor $S_b(\mathcal{R}) > 0$ for every bounded receiver $\mathcal{R}$, and receiver-decoded coherence. The image of $S_{\mathcal{R}}$ lies in $[S_b(\mathcal{R}), 100]$.

A key consequence—the *Unconstrained Subtask Theorem* of ?—states that any S-value S^*

admits an arbitrary type-mixed decomposition into sub-expressions, signed or fractional, provided their composition under the receiver's evaluation map yields S^* . This freedom is what permits heterogeneous monetary derivatives to be decomposed and recombined without constraint.

3 The Transaction Clock

Definition 3.1 (Transaction clock). Let $G(\cdot, \tau)$ be the gain-loss process of (??). The *transaction clock* is the right-continuous, non-decreasing process

$$\Theta(t) := \int_0^t |G(s, \tau)| ds, \quad t \geq 0. \quad (4)$$

The transaction clock accumulates the absolute magnitude of economic activity. When activity is high, Θ advances rapidly; when markets are inactive, Θ nearly stalls.

Lemma 3.2 (Basic properties of Θ). *Under the standing assumptions:*

- (i) $\Theta(0) = 0$ and Θ is $(\mathcal{F}_t)$ -adapted;
- (ii) Θ is non-decreasing and absolutely continuous with $\dot{\Theta}(t) = |G(t, \tau)|$ a.e.;
- (iii) Θ is a finite-variation process with quadratic variation $[\Theta]_t = 0$;
- (iv) if $\mathbb{E}[|G(t, \tau)|] < \infty$ for all t , then $\mathbb{E}[\Theta(t)] < \infty$.

Proof. (i) follows from G being adapted and the integral being a progressively measurable operation. (ii) is immediate from the fundamental theorem of calculus for absolutely continuous functions. (iii) follows because Θ has bounded variation, so its quadratic variation vanishes. (iv) follows by Fubini's theorem: $\mathbb{E}[\Theta(t)] = \int_0^t \mathbb{E}[|G(s, \tau)|] ds < \infty$. $\square$

Definition 3.3 (Transaction-time inverse). The *transaction-time inverse* is the right-continuous process

$$\theta(s) := \inf\{t \geq 0 : \Theta(t) > s\}, \quad s \geq 0. \quad (5)$$

The pair (Θ, θ) implements a random time change in the sense of Clark (?): a process X observed in calendar time is transformed to $X_{\theta(s)}$ observed in transaction time s .

Proposition 3.4 (Subordination). *Let $X(t)$ be a continuous semimartingale. Then $Y(s) := X(\theta(s))$ is a process in transaction time whose infinitesimal increments satisfy*

$$dY(s) = dX(\theta(s)) \cdot \frac{1}{|G(\theta(s), \tau)|} ds. \quad (6)$$

In particular, $dY(s)/ds$ is the rate of change of X per unit of transaction activity.

Proof. By the chain rule for absolutely continuous time changes (?), since Θ is absolutely continuous with density $|G|$, $d\theta(s)/ds = 1/|G(\theta(s))|$. Therefore $dY(s) = (dX/dt)|_{t=\theta(s)} \cdot (d\theta/ds) ds = (dX/dt)/|G| ds$ as claimed. $\square$

Remark 3.5. Proposition ?? generalises Clark's result from trade-count subordination to gain-loss subordination. The economic interpretation is that one unit of transaction time represents one unit of cumulative gain-loss magnitude, making the clock a direct measure of economic intensity rather than a proxy (volume, trade count).

Theorem 3.6 (Variance rescaling under subordination). *Let $X(t)$ be a zero-mean martingale with $\text{Var}[X(t)] = \sigma^2 t$ and let $Y(s) = X(\theta(s))$. Then*

$$\text{Var}[Y(s)] = \sigma^2 \mathbb{E}[\theta(s)]. \quad (7)$$

Under the ergodic assumption $\mathbb{E}[|G(t)|] \rightarrow \bar{g} > 0$ as $t \rightarrow \infty$, $\mathbb{E}[\theta(s)] \sim s/\bar{g}$ and $\text{Var}[Y(s)] \sim \sigma^2 s/\bar{g}$.

Proof. By the optional stopping theorem and the tower property, $\mathbb{E}[X(\theta(s))^2] = \sigma^2 \mathbb{E}[\theta(s)]$, which gives (7). The ergodic estimate follows from the law of large numbers: $\Theta(t)/t \rightarrow \bar{g}$ a.s. implies $\theta(s)/s \rightarrow 1/\bar{g}$ a.s. by the inverse function theorem for monotone processes, so $\mathbb{E}[\theta(s)] \rightarrow s/\bar{g}$ by dominated convergence under integrability conditions. $\square$

4 The Monetary Derivative Operator

Definition 4.1 (Monetary derivative). Let $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ be a differentiable function of calendar time. Its *monetary derivative* at time t (with respect to the transaction clock Θ) is

$$\frac{df}{d\Theta}(t) := \frac{df/dt}{|G(t, \tau)|} = \frac{\dot{f}(t)}{|G(t, \tau)|} \quad (8)$$

whenever $G(t, \tau) \neq 0$.

Remark 4.2. If f is measured in units $[f]$ and calendar time is measured in seconds, then df/dt has units $[f] \cdot s^{-1}$. The gain-loss $|G|$ is a dimensionless portfolio return, hence $\frac{df}{d\Theta}$ has units $[f]$ per unit return—a quantity that still depends on the measurement scale of f . Section ?? resolves this residual dimensionality by passing to S-entropy coordinates.

Lemma 4.3 (Linearity). *For differentiable f, g and constants $\alpha, \beta \in \mathbb{R}$:*

$$\frac{d(\alpha f + \beta g)}{d\Theta} = \alpha \frac{df}{d\Theta} + \beta \frac{dg}{d\Theta}. \quad (9)$$

Proof. $d(\alpha f + \beta g)/dt = \alpha \dot{f} + \beta \dot{g}$; divide by $|G| \neq 0$. $\square$

Lemma 4.4 (Product rule). *For differentiable f, g :*

$$\frac{d(fg)}{d\Theta} = g \frac{df}{d\Theta} + f \frac{dg}{d\Theta}. \quad (10)$$

Proof. $(fg)' = \dot{f}g + f\dot{g}$; divide by $|G|$ and use Definition ??.

Lemma 4.5 (Chain rule). *Let $f = h \circ g$ where $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable. Then*

$$\frac{df}{d\Theta}(t) = h'(g(t)) \frac{dg}{d\Theta}(t). \quad (11)$$

Proof. $\dot{f} = h'(g)\dot{g}$; divide by $|G|$. $\square$

Proposition 4.6 (Monetary derivative in transaction time). *Let $Y(s) = f(\theta(s))$ be f observed in transaction time. Then*

$$\frac{dY}{ds}(s) = \frac{df}{d\Theta}(\theta(s)). \quad (12)$$

That is, the monetary derivative in calendar time coincides with the ordinary derivative in transaction time.

Proof. $dY/ds = (df/dt)|_{t=\theta(s)} \cdot d\theta/ds = \dot{f}(\theta(s))/|G(\theta(s))| = \frac{df}{d\Theta}(\theta(s))$. $\square$

Remark 4.7. Proposition ?? shows that working with monetary derivatives in calendar time is *identical* to working with ordinary derivatives in transaction time. The two representations are related by the random time change (Θ, θ) and are freely interchangeable.

Theorem 4.8 (Bounded activity bound). *Suppose $|\dot{f}(t)| \leq C$ for all t and $|G(t)| \geq g_{\min} > 0$ a.s. for all t . Then*

$$\left| \frac{df}{d\Theta}(t) \right| \leq \frac{C}{g_{\min}} \quad \text{a.s.} \quad (13)$$

Proof. $|\frac{df}{d\Theta}| = |\dot{f}|/|G| \leq C/g_{\min}$. $\square$

Theorem 4.9 (Monetary fundamental theorem). *For any differentiable f with integrable $\frac{df}{d\Theta}$:*

$$f(t_2) - f(t_1) = \int_{\Theta(t_1)}^{\Theta(t_2)} \frac{df}{d\Theta}(\theta(s)) ds. \quad (14)$$

Proof. By the change-of-variables formula for monotone increasing Θ : $\int_{t_1}^{t_2} \dot{f}(t) dt = \int_{\Theta(t_1)}^{\Theta(t_2)} \dot{f}(\theta(s)) \frac{d\theta}{ds} ds = \int_{\Theta(t_1)}^{\Theta(t_2)} \frac{\dot{f}(\theta(s))}{|G(\theta(s))|} ds$. The result follows from $f(t_2) - f(t_1) = \int_{t_1}^{t_2} \dot{f}(t) dt$. $\square$

5 S-Entropy as Dimensionless Currency

The monetary derivative $\frac{df}{d\Theta}$ in Definition ?? has residual units $[f]$ inherited from f . Comparing monetary derivatives of price (in currency units) with those of volume (in share counts) or volatility (dimensionless) requires a common unit. The S-entropy functional (??) provides exactly this.

Definition 5.1 (S-transformed quantity). Let $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ be a quantity of interest with truth value $f^*(t)$ at time t , and let $\mathcal{R}$ be a fixed receiver. The *S-transformed quantity* is

$$\tilde{f}(t) := S_{\mathcal{R}}(f(t), f^*(t)) \in [0, 100]. \quad (15)$$

Proposition 5.2 (Dimensionlessness of S-transform). *For any quantities f and g measured in different units, the S-transforms $\tilde{f}$ and $\tilde{g}$ are both measured in S-units on the scale $[0, 100]$, and their monetary derivatives $\frac{d\tilde{f}}{d\Theta}$ and $\frac{d\tilde{g}}{d\Theta}$ are both in S-units per unit transaction magnitude—the same unit—making them directly addable.*

Proof. $\tilde{f}$ and $\tilde{g}$ are both images of the S-functional, which maps into $[0, 100] \subset \mathbb{R}$ with no physical dimension. The transaction clock Θ accumulates the dimensionless portfolio return $|G|$, so Θ is also dimensionless. Therefore $d\tilde{f}/d\Theta$ and $d\tilde{g}/d\Theta$ are both dimensionless ratios in $\mathbb{R}$. $\square$

Corollary 5.3 (Additivity across heterogeneous quantities). *For any finite collection of quantities $f_1, \dots, f_n$ measured in arbitrary units and any weights $\alpha_i \in \mathbb{R}$ with $\sum_i |\alpha_i| < \infty$:*

$$\sum_{i=1}^n \alpha_i \frac{d\tilde{f}_i}{d\Theta}(t) \in \mathbb{R} \quad (16)$$

is well-defined and carries the single interpretation of a net monetary derivative in S-units per transaction unit.

Proof. Each $\frac{d\tilde{f}_i}{d\Theta}$ is a real number by Proposition ??, and a finite linear combination of real numbers is a real number. $\square$

Remark 5.4. Corollary ?? formalises the informal observation that S-entropy allows one to *add apples and oranges*: by mapping every quantity into the common currency $[0, 100]$ before differentiating, the monetary derivative strips away all legacy units and returns a universal scalar. The Unconstrained Subtask Theorem of ? further guarantees that any such sum can be decomposed into sub-tasks in an arbitrary type-mixed fashion without violating the total S-value.

Theorem 5.5 (Floor persistence under monetary differentiation). *Let $\mathcal{R}$ be a bounded receiver with floor $S_b(\mathcal{R}) > 0$. Then for every quantity f with $\tilde{f}(t) \geq S_b(\mathcal{R})$ for all t and every interval $[t_1, t_2]$:*

$$\tilde{f}(t_2) - \tilde{f}(t_1) = \int_{\Theta(t_1)}^{\Theta(t_2)} \frac{d\tilde{f}}{d\Theta}(\theta(s)) ds, \quad (17)$$

and the integral is bounded: $|\tilde{f}(t_2) - \tilde{f}(t_1)| \leq 100 - S_b(\mathcal{R})$.

Proof. The identity follows from Theorem ?? applied to $\tilde{f}$. The bound follows from $\tilde{f} \in [S_b(\mathcal{R}), 100]$, so $|\tilde{f}(t_2) - \tilde{f}(t_1)| \leq 100 - S_b(\mathcal{R})$. $\square$

6 The Monetary Tangent Space

Definition 6.1 (Monetary tangent space). Fix a receiver $\mathcal{R}$ and a transaction-time point $s \geq 0$. The *monetary tangent space at s* is the vector space

$$\mathcal{T}_s^\Theta := \left\{ v = \frac{d\tilde{f}}{d\Theta}(\theta(s)) \mid f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R} \text{ differentiable, } \tilde{f} \text{ as in Definition ??} \right\} \subseteq \mathbb{R}^n \quad (18)$$

Remark 6.2. Since all S-transformed quantities take values in $[0, 100]$, the monetary tangent space $\mathcal{T}_s^\Theta$ is a subspace of $\mathbb{R}$. In the finite-dimensional case with n economic observables, the monetary tangent space is $\mathcal{T}_s^\Theta \subseteq \mathbb{R}^n$, one coordinate per observable.

Proposition 6.3 (Monetary tangent space is a normed space). *Define the monetary norm on $\mathbb{R}^n$ by*

$$\|v\|_\Theta := \|v\|_2 \cdot \frac{1}{\sqrt{n}} \cdot \frac{1}{100}, \quad v \in \mathbb{R}^n. \quad (19)$$

The pair $(\mathbb{R}^n, \|\cdot\|_\Theta)$ is a normed vector space, and $\mathcal{T}_s^\Theta \subseteq \mathbb{R}^n$ inherits this norm.

Proof. The monetary norm is a scalar multiple of the Euclidean norm, hence satisfies positivity, absolute homogeneity, and the triangle inequality. The prefactor $(100\sqrt{n})^{-1}$ normalises the range so that the maximum possible monetary tangent vector $(100, \dots, 100)/100\sqrt{n}$ has monetary norm one. $\square$

Theorem 6.4 (Closure under composition). *Let $f_1, \dots, f_n$ be S-transformed differentiable quantities and let $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. Then $\tilde{h} := S_{\mathcal{R}}(\Phi(f_1, \dots, f_n), \Phi(f_1^*, \dots, f_n^*))$ satisfies*

$$\frac{d\tilde{h}}{d\Theta} = \sum_{i=1}^n \frac{\partial \Phi}{\partial \tilde{f}_i} \cdot \frac{d\tilde{f}_i}{d\Theta}, \quad (20)$$

and $\frac{d\tilde{h}}{d\Theta} \in \mathcal{T}_s^\Theta$.

Proof. By the chain rule (Lemma ??) applied to $\tilde{h} = S_{\mathcal{R}}(\Phi(\cdot), \cdot)$ and the multivariate chain rule for Φ : $\dot{\tilde{h}} = \sum_i \partial \Phi / \partial \tilde{f}_i \cdot \dot{\tilde{f}}_i$. Dividing by $|G|$ gives the monetary derivative formula. Since each $\frac{d\tilde{f}_i}{d\Theta} \in \mathcal{T}_s^\Theta$ and $\mathcal{T}_s^\Theta$ is closed under finite linear combinations, the sum lies in $\mathcal{T}_s^\Theta$. $\square$

Definition 6.5 (Monetary gradient). For a scalar function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ of n economic observables, the *monetary gradient* is

$$\nabla_\Theta \Phi := \left(\frac{d(\partial \Phi / \partial \tilde{f}_i)}{d\Theta} \right)_{i=1}^n \in \mathcal{T}_s^{\Theta n}. \quad (21)$$

Corollary 6.6 (Monetary gradient descent). *A quantity Φ evolves toward its truth value Φ^* in transaction time at rate*

$$\frac{d\tilde{\Phi}}{ds}(s) = -\|\nabla_\Theta \Phi\|_\Theta^2 \cdot \frac{1}{\frac{d\tilde{\Phi}}{ds}}(\theta(s)) \quad (22)$$

along the monetary gradient flow $\dot{\tilde{f}}_i = -\partial \Phi / \partial \tilde{f}_i$.

Proof. Along the gradient flow, $d\tilde{\Phi}/dt = \sum_i (\partial\Phi/\partial\tilde{f}_i) \cdot \dot{\tilde{f}}_i = -\|\nabla\Phi\|^2$. Dividing by $|G|$ gives $\frac{d\tilde{\Phi}}{d\Theta} = -\|\nabla\Phi\|^2/|G|$. In transaction time this becomes $d\tilde{\Phi}/ds = \frac{d\tilde{\Phi}}{d\Theta}(\theta(s))$. $\square$

7 Hierarchical Gear Networks and Telescoping Gear-Ratio Derivatives

We now develop the interaction between the monetary derivative and the hierarchical gear network of nested stopping times introduced in the context of portfolio rebalancing.

Definition 7.1 (Gear network). A K -layer gear network with thresholds $\theta_1 < \theta_2 < \dots < \theta_K$ is a sequence of stopping times defined recursively as follows. Set $I_0(t) := G(t, \tau)$ (the base-layer gain-loss). For each $k = 1, \dots, K$:

- (i) Layer- k transaction clock: $\Theta_k(t) := \int_0^t |I_{k-1}(s)| ds$;
- (ii) Layer- k stopping time: $\sigma_k := \inf\{t \geq 0 : \Theta_k(t) > \theta_k\}$;
- (iii) Layer- k accumulated imbalance: $I_k(t) := \Theta_k(t) - \Theta_k(\sigma_k^-)$ (reset to zero after each trigger).

Remark 7.2. Definition ?? defines a cascade: layer-1 fires when cumulative gain-loss exceeds θ_1 ; layer-2 fires when cumulative layer-1 activity exceeds θ_2 ; and so on. The ratio θ_{k+1}/θ_k is the gear ratio between layers k and $k+1$.

Theorem 7.3 (Gear hierarchy mean crossing times). *Under the martingale difference structure and the independence of increments across reset epochs, the expected number of layer- $(k+1)$ firings in $[0, T]$ satisfies*

$$\mathbb{E}[N_{k+1}(T)] = \frac{\theta_k}{\theta_{k+1}} \cdot \mathbb{E}[N_k(T)] \cdot (1 + o(1)) \quad (23)$$

as $T \rightarrow \infty$, so that layer- k fires approximately θ_{k+1}/θ_k times per layer- $(k+1)$ firing.

Proof. Between consecutive layer- $(k+1)$ firings, layer- k must accumulate activity θ_{k+1} . Each individual layer- k firing contributes exactly θ_k of activity (by definition of the stopping time), so the expected number of layer- k firings required is θ_{k+1}/θ_k . By the renewal argument (?) applied to the i.i.d. inter-arrival times of

layer- k firings (guaranteed by the reset structure and the martingale difference property), $\mathbb{E}[N_{k+1}(T)] \sim \mathbb{E}[N_k(T)] \cdot (\theta_k/\theta_{k+1})$. $\square$

Definition 7.4 (Layer- k monetary derivative). For a quantity f and layer $k \in \{1, \dots, K\}$, the layer- k monetary derivative is

$$\frac{df}{d\Theta_k}(t) := \frac{df/dt}{|\Theta_k(t)|/t} = \frac{t \dot{f}(t)}{|\Theta_k(t)|}. \quad (24)$$

Theorem 7.5 (Telescoping gear-ratio formula). *For any differentiable quantity f and K -layer gear network:*

$$\frac{d\tilde{f}}{d\Theta_K}(t) = \frac{d\tilde{f}}{d\Theta}(t) \cdot \prod_{k=1}^{K-1} \rho_k(t), \quad (25)$$

where $\rho_k(t) := \Theta_k(t)/\Theta_{k+1}(t)$ is the instantaneous gear ratio at layer k .

Proof. Write $\frac{d\tilde{f}}{d\Theta_K} = \dot{\tilde{f}}/|\Theta_K/t| = (\dot{\tilde{f}}/|\Theta_1/t|) \cdot (|\Theta_1/t|/|\Theta_K/t|)$. By telescoping, $|\Theta_1/t|/|\Theta_K/t| = \prod_{k=1}^{K-1} (|\Theta_k/t|/|\Theta_{k+1}/t|) = \prod_{k=1}^{K-1} \rho_k(t)$. Since $\frac{d\tilde{f}}{d\Theta} = \dot{\tilde{f}}/|\Theta_1/t|$ (identifying the base layer with $k = 1$), the result follows. $\square$

Corollary 7.6 (Gear amplification and attenuation). *If thresholds satisfy $\theta_1 < \theta_2 < \dots < \theta_K$, then $\Theta_k(t) < \Theta_{k+1}(t)$ for all t in the long-run mean, so $\rho_k(t) < 1$ for each k , and the layer- K monetary derivative is attenuated relative to the base layer: $|\frac{d\tilde{f}}{d\Theta_K}| \leq |\frac{d\tilde{f}}{d\Theta}|$. Conversely, decreasing thresholds produce amplification.*

Proof. By Theorem ??, higher layers fire less frequently, hence their transaction clocks advance more slowly: $\Theta_{k+1}(t) \geq \Theta_k(t)$ in expectation. This gives $\rho_k(t) \leq 1$. $\square$

Theorem 7.7 (Gear-ratio stability under ergodic dynamics). *Under a stationary ergodic gain-loss process with $\mathbb{E}[|G(t)|] = \bar{g} > 0$, the instantaneous gear ratios converge almost surely:*

$$\rho_k(t) \xrightarrow[t \rightarrow \infty]{a.s.} \bar{\rho}_k := \frac{\bar{g}_k}{\bar{g}_{k+1}}, \quad (26)$$

where $\bar{g}_k := \mathbb{E}[|I_{k-1}(t)|]$ is the stationary mean activity of layer k . Therefore, the telescoping product converges to a deterministic limit $\prod_{k=1}^{K-1} \bar{\rho}_k$.

Proof. By the ergodic theorem (?), $\Theta_k(t)/t \rightarrow \bar{g}_k$ a.s. for each k . Therefore $\rho_k(t) = \Theta_k(t)/\Theta_{k+1}(t) \rightarrow \bar{g}_k/\bar{g}_{k+1} = \bar{\rho}_k$ a.s. The product of $K-1$ almost surely convergent sequences converges almost surely to their product. $\square$

8 The No-Privileged-Level Theorem in Transaction Time

A central result of ? is the *Recursive No-Privileged-Level Theorem*: every S-value decomposes into a triple of S-values at the next finer depth, with the same algebraic laws holding at every depth. We now prove that this principle is preserved exactly under the monetary derivative operation.

Theorem 8.1 (No-Privileged-Level in transaction time). *Let $k, \ell \in \{1, \dots, K\}$ be any two gear layers. The algebraic laws of the monetary derivative at layer k are identical to those at layer ℓ : specifically,*

- (i) Linearity: $\frac{d(\alpha\tilde{f} + \beta\tilde{g})}{d\Theta_k} = \alpha \frac{d\tilde{f}}{d\Theta_k} + \beta \frac{d\tilde{g}}{d\Theta_k}$;
- (ii) Product rule: $\frac{d(\tilde{f}\tilde{g})}{d\Theta_k} = \tilde{g} \frac{d\tilde{f}}{d\Theta_k} + \tilde{f} \frac{d\tilde{g}}{d\Theta_k}$;
- (iii) Chain rule: $\frac{d(h \circ \tilde{f})}{d\Theta_k} = h'(\tilde{f}) \frac{d\tilde{f}}{d\Theta_k}$;
- (iv) Fundamental theorem: $\tilde{f}(t_2) - \tilde{f}(t_1) = \int_{\Theta_k(t_1)}^{\Theta_k(t_2)} \frac{d\tilde{f}}{d\Theta_k}(\theta_k(s)) ds$.

These laws hold identically for every $k \in \{1, \dots, K\}$.

Proof. Each law (i)–(iv) is proved by dividing the corresponding calendar-time identity by $|\Theta_k(t)|/t$, which appears as a common factor and cancels. The resulting identities are identical in form for all k , since the specific layer enters only through the denominator Θ_k , and all the algebraic manipulations (linearity, Leibniz rule, chain rule, FTC) proceed identically regardless of which layer’s clock is used. $\square$

Corollary 8.2 (Level-independent calculus). *The set of monetary derivatives $\{\frac{d\tilde{f}}{d\Theta_k}\}_{k=1}^K$ forms a family of calculi indexed by gear layer, all sharing the same algebraic structure. No layer carries greater algebraic privilege than any other.*

Remark 8.3. Theorem ?? is the analogue in transaction time of the Recursive No-Privileged-Level Theorem of ?: just as every S-value is its own sub-problem at the next depth (with the

same laws), every gear layer defines an equivalent calculus of monetary derivatives. The gear ratios ρ_k encode the *quantitative* differences between layers (the amplification or attenuation of rate signals) without introducing any *qualitative* algebraic asymmetry.

Theorem 8.4 (S-entropy floor preservation across layers). *For every bounded receiver $\mathcal{R}$ with floor $S_b(\mathcal{R}) > 0$ and every gear layer k :*

$$\tilde{f}(t) \in [S_b(\mathcal{R}), 100] \quad \forall t \geq 0, \quad (27)$$

and this constraint propagates layer by layer: $\frac{d\tilde{f}}{d\Theta_k}$ is bounded by $(100 - S_b(\mathcal{R}))/\Theta_{k,\min}$ whenever $\Theta_k(t) \geq \Theta_{k,\min} > 0$.

Proof. $\tilde{f}(t) \in [S_b(\mathcal{R}), 100]$ by Axioms (S1)–(S3) of the S-functional. Therefore $|\dot{\tilde{f}}(t)| \leq 100 - S_b(\mathcal{R})$ almost everywhere, since $\tilde{f}$ cannot leave the bounded interval. Dividing by $|\Theta_k(t)|/t \geq \Theta_{k,\min}/t$ gives $|\frac{d\tilde{f}}{d\Theta_k}| \leq t(100 - S_b)/|\Theta_k(t)| \leq (100 - S_b)/\Theta_{k,\min}$. $\square$

9 Ergodic Theory of Monetary Derivatives

Theorem 9.1 (Ergodic consistency of monetary derivatives). *Let the return process $(R(t, \tau))_{t \geq 0}$ be stationary and ergodic under $\mathbb{P}$, with invariant measure π . Let f be a measurable function of the returns with $\mathbb{E}_\pi[|\dot{f}|/\bar{g}] < \infty$. Then*

$$\frac{1}{S} \int_0^S \frac{d\tilde{f}}{d\Theta}(\theta(s)) ds \xrightarrow[S \rightarrow \infty]{a.s.} \mathbb{E}_\pi \left[\frac{d\tilde{f}}{d\Theta} \right] = \frac{\mathbb{E}_\pi[\dot{f}]}{\bar{g}}. \quad (28)$$

Proof. By Proposition ??, $\frac{d\tilde{f}}{d\Theta}(\theta(s)) = dY/ds(s)$ where $Y(s) = \tilde{f}(\theta(s))$. The time-average on the left of (??) is the Birkhoff average of dY/ds in transaction time. Since the return process is ergodic and Θ subordinates the process to the transaction time, $Y(s)$ inherits ergodicity from the underlying stationary process (see ?, Theorem 4.1.2). Birkhoff’s ergodic theorem then gives $S^{-1} \int_0^S (dY/ds) ds \rightarrow \mathbb{E}_\pi[dY/ds]$ a.s. The expectation equals $\mathbb{E}_\pi[\dot{f}]/\bar{g}$ by the ergodic theorem applied to $\dot{f}$ in calendar time, combined with $\Theta(t)/t \rightarrow \bar{g}$ a.s. $\square$

Theorem 9.2 (Convergence rate of the Birkhoff average). *Under the conditions of Theorem ??*

and additionally assuming $\mathbb{E}_\pi[(\dot{f}/\bar{g})^2] < \infty$ and a finite mixing time:

$$\left\| \frac{1}{S} \int_0^S \frac{d\tilde{f}}{d\Theta}(\theta(s)) ds - \mathbb{E}_\pi \left[\frac{d\tilde{f}}{d\Theta} \right] \right\|_2 = O(S^{-1/2}). \quad (29)$$

Proof. Under the mixing condition, the central limit theorem for ergodic processes (?) applies to the partial sums $\sum_{k=1}^K (dY/ds)|_{s=k\Delta}$. In the continuous-time limit this gives the $O(S^{-1/2})$ rate of convergence, consistent with the Birkhoff theorem applied to a square-integrable function. $\square$

Corollary 9.3 (Monetary law of large numbers). *For a discrete sequence of transaction-time observations $s_1 < s_2 < \dots < s_K$ with spacing Δs :*

$$\frac{1}{K} \sum_{k=1}^K \frac{d\tilde{f}}{d\Theta}(\theta(s_k)) \xrightarrow[K \rightarrow \infty]{a.s.} \mathbb{E}_\pi \left[\frac{d\tilde{f}}{d\Theta} \right]. \quad (30)$$

Proof. Apply Theorem ?? to the discrete-time Birkhoff average; ergodicity of the discrete skeleton follows from that of the continuous process (?). $\square$

Theorem 9.4 (Inter-layer ergodic consistency). *Under the same ergodicity conditions, for any two layers $j, k \in \{1, \dots, K\}$:*

$$\frac{\mathbb{E}_\pi[\frac{d\tilde{f}}{d\Theta_k}]}{\mathbb{E}_\pi[\frac{d\tilde{f}}{d\Theta_j}]} = \frac{\bar{g}_j}{\bar{g}_k}, \quad (31)$$

and the ratio is determined solely by the mean activity rates $\bar{g}_j, \bar{g}_k$ of the respective layers.

Proof. $\mathbb{E}_\pi[\frac{d\tilde{f}}{d\Theta_k}] = \mathbb{E}_\pi[\dot{f}]/\bar{g}_k$ by Theorem ?? applied at layer k , and similarly for layer j . Dividing gives the ratio. $\square$

10 Worked Examples

10.1 Price velocity in transaction time

Let $P(t)$ be the mid-price of a single asset with truth value $P^*(t)$ = fundamental value. The S-transform is $\tilde{P}(t) = S_{\mathcal{R}}(P(t), P^*(t))$. Suppose prices move at rate $\dot{P}(t)$ and the portfolio

generates gain-loss $G(t)$. Then the monetary price velocity is

$$\frac{d\tilde{P}}{d\Theta}(t) = \frac{dS(P, P^*)/dt}{|G(t)|} = \frac{\partial S / \partial P \cdot \dot{P}}{|G|}. \quad (32)$$

This answers the question: how fast is the price diverging from fundamental per unit of economic activity? A large $\frac{d\tilde{P}}{d\Theta}$ means rapid mispricing per transaction unit; a small value means the mispricing is moving slowly relative to the economic intensity of the market.

10.2 Cross-asset comparison: price and volume

Consider two assets with price P and volume V . Their individual monetary derivatives $\frac{d\tilde{P}}{d\Theta}$ and $\frac{d\tilde{V}}{d\Theta}$ are both in S-units per transaction unit. Their sum

$$\frac{d\tilde{P}}{d\Theta} + \frac{d\tilde{V}}{d\Theta} \quad (33)$$

is a single dimensionless number representing the *total rate of informational divergence* of both price and volume from their respective truths, per unit of transactional activity. This would be meaningless in calendar time (dollars-per-second plus shares-per-second is ill-defined); in transaction time with S-entropy normalisation, it is well-defined by Corollary ??.

10.3 Two-layer gear derivative of volatility

Let $\sigma(t)^2$ be the realised variance of a portfolio. Using a two-layer gear network with thresholds $\theta_1 = 0.1$ and $\theta_2 = 0.5$:

$$\frac{d\tilde{\sigma}}{d\Theta_2}(t) = \frac{d\tilde{\sigma}}{d\Theta}(t) \cdot \rho_1(t) = \frac{\dot{\sigma}}{|G|} \cdot \frac{\Theta_1(t)}{\Theta_2(t)}. \quad (34)$$

The factor $\rho_1(t) = \Theta_1(t)/\Theta_2(t) < 1$ attenuates the base-layer signal: the layer-2 monetary derivative of volatility represents the rate at which volatility deviates from truth *only as reckoned by the slow macro-rebalancing clock*, filtering out high-frequency noise from the base-layer activity.

11 Implications and Open Questions

11.1 Relation to dimensional analysis

The monetary derivative in S-units is the first *automatically dimensionless* rate operator for economic systems. Classical dimensional analysis (?) requires explicit unit-tracking; the Buckingham Π theorem (?) constructs dimensionless groups post-hoc. The monetary derivative achieves dimensionlessness directly, by using S-entropy as the universal scalar field and accumulated absolute gain-loss as the universal clock.

11.2 Connection to information geometry

The monetary tangent space $\mathcal{T}_s^\Theta$ with the monetary norm $\|\cdot\|_\Theta$ is a Riemannian manifold when the S-functional varies smoothly with the underlying process. The monetary gradient of Definition ?? defines a natural gradient flow on this manifold. The connection to information geometry (?) and the Fisher information metric is an open question.

11.3 Stochastic monetary calculus

When $G(t)$ is stochastic, the monetary derivative $\frac{df}{d\Theta} = \dot{f}/|G|$ is a random variable. A full stochastic calculus in transaction time—including Itô’s formula for $d\tilde{f}$ in $d\Theta$ —follows from the theory of stochastic time changes (?), but the S-entropy nonlinearity requires careful treatment. We leave the development of a complete Itô calculus for S-entropy-valued processes as an open question.

11.4 Gear networks with variable thresholds

Theorem ?? and the No-Privileged-Level result (Theorem ??) assume fixed thresholds θ_k . Adaptive threshold sequences $\theta_k(t)$ that respond to realised volatility would make the gear ratios $\rho_k(t)$ adaptive as well. Whether the No-Privileged-Level structure is preserved under such adaptations is an open question with practical significance for adaptive rebalancing algorithms.

11.5 Summary

We have constructed a self-consistent calculus in which:

1. the fundamental rate operator is the monetary derivative $d/d\Theta$, measuring change per unit of accumulated gain-loss intensity;
2. S-entropy provides a universal, dimensionless scalar field that makes rates of change of heterogeneous quantities addable;
3. the monetary tangent space is a normed vector space closed under the monetary derivative;
4. a telescoping gear-ratio formula propagates monetary derivatives through any finite gear network;
5. the No-Privileged-Level theorem ensures that the same algebraic laws hold at every gear layer; and
6. under stationary ergodic dynamics, time-averaged monetary derivatives converge almost surely at rate $O(K^{-1/2})$.

The calculus reduces to standard calculus when $|G|$ is constant, and subsumes Clark’s subordination framework as the special case where S-entropy is replaced by the identity function.